

Title: Machine Learning Framework for Predictive Maintenance of Industrial Equipment Using Sensor Data

BATCH: 06

Abstract

Industrial equipment requires consistent operation to maintain productivity and reduce operational costs. Traditional maintenance strategies, such as reactive or time-based preventive maintenance, often lead to unexpected catastrophic failures or unnecessary servicing, providing limited insight into real-time mechanical health. This project proposes a Machine Learning Framework for Predictive Maintenance (PdM) of Industrial Equipment Using Sensor Data that leverages multi-modal sensory inputs to continuously monitor equipment health, detect early signs of mechanical degradation, and provide actionable maintenance alerts.

The proposed system uses advanced Machine Learning architectures such as Random Forest, XGBoost, or Long Short-Term Memory (LSTM) networks and trains them on benchmark predictive maintenance datasets. The system performs data preprocessing, feature engineering (analyzing vibration, temperature, and pressure signals), anomaly detection, and Remaining Useful Life (RUL) forecasting. To improve interpretability, feature importance techniques and explainable AI methods (such as SHAP) are incorporated to identify the specific sensor parameters that contribute most significantly to a predicted mechanical failure. The framework can classify equipment states into categories such as normal operation, early warning, and critical failure risk, depending on the selected dataset and operational thresholds.

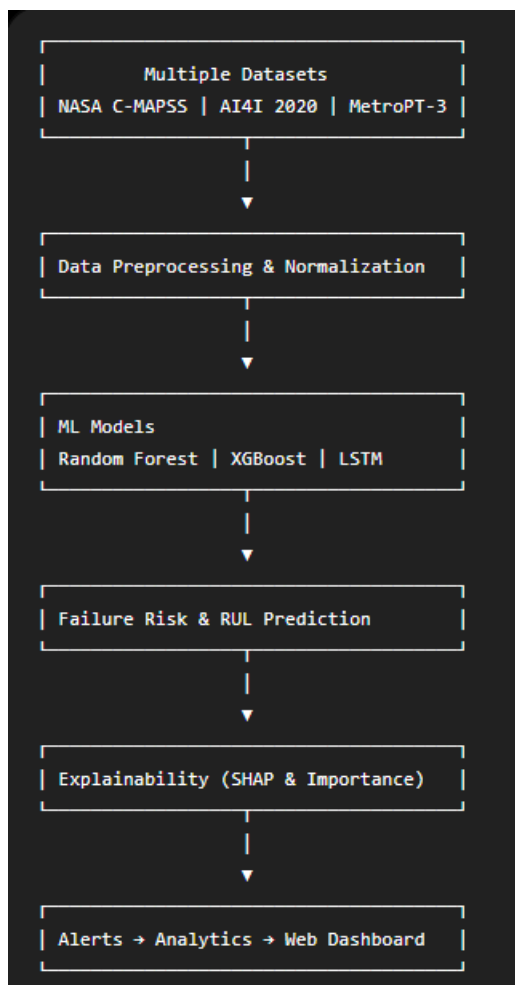
The proposed approach can further analyze equipment data at an aggregate level to identify dominant failure modes, recurring mechanical issues, degradation patterns, and areas requiring immediate maintenance attention. The performance of the predictive models is evaluated using accuracy, precision, recall, F1 score, and confusion matrix for classification tasks, as well as Root Mean Square Error (RMSE) for RUL prediction. The resulting system aims to provide not only accurate predictive maintenance analysis but also transparent and actionable insights that can support optimized maintenance scheduling, reduced unplanned downtime, and improved operational decision-making.

To ensure practical applicability in real-world manufacturing environments, the framework is designed for seamless integration with existing Industrial Internet of Things (IIoT) infrastructure. By deploying the trained models on edge computing devices or cloud-based platforms, the system facilitates continuous, low-latency processing of streaming sensor data. End-users can interact with the system through an intuitive web-based dashboard that visualizes equipment health trajectories, highlights critical sensor anomalies in real time, and issues automated alerts. Ultimately, this scalable solution not only extends asset longevity but also supports broader industrial sustainability goals by minimizing the resource waste and energy consumption typically associated with faulty machinery.

Datasets:

Dataset	URL link
NASA C-MAPSS (Turbofan Engine Degradation)	NASA Prognostics Data Repository
AI4I 2020 Predictive Maintenance Dataset	UCI Machine Learning Repository
MetroPT-3 Dataset (Predictive Maintenance)	UCI Machine Learning Repository

Approach:



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